INCH-POUND
MIL-PRF-3098/84F
6 March 2013
SUPERSEDING
MIL-PRF-3098/84E
16 January 2006

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR108/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 50 MHz to 93 MHz; fifth mechanical overtone; controlled; series resonance.

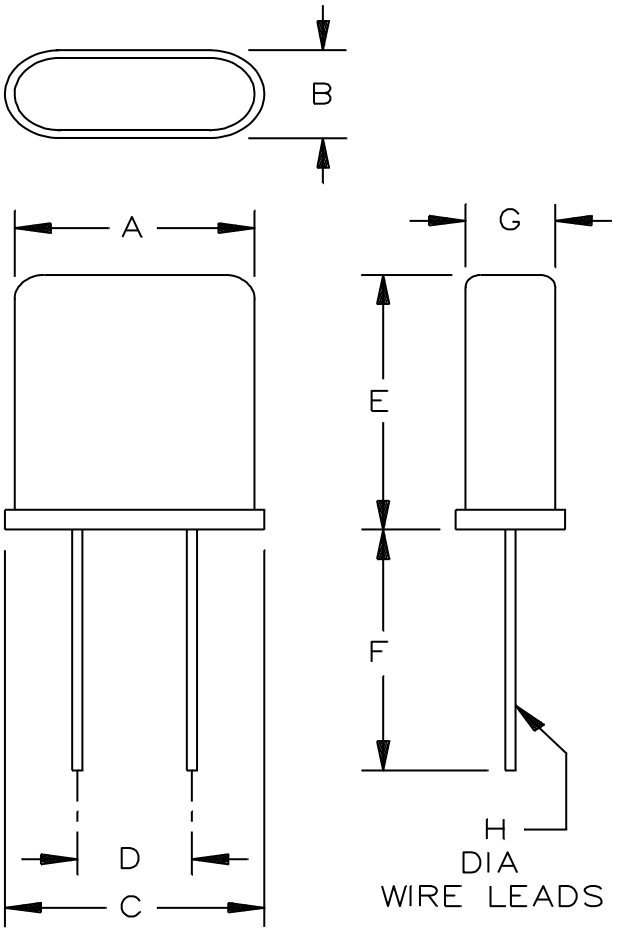


FIGURE 1. Crystal unit - CR108/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.015	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR108/U, crystal unit - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1. 1/

Frequency range: 50 MHz to 93 MHz, inclusive.

Capacitance, shunt: 5.5 pF, maximum.

Frequency stability: ±2.5 ppm.

Mode of oscillation: Fifth mechanical overtone.

Operating temperature range (controlled): +80°C to +90°C, inclusive.

Frequency tolerance: ±10 ppm (including frequency stability).

Equivalent resistance: 50 ohms, maximum.

Operable temperature range: -55°C to +80°C and +90°C to +105°C.

Reference temperature: +85°C ±1°C.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration high frequency: Method 204, MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±3 ppm.

1/ These crystal units were previously fabricated using Air Force Technical Order 12-1-20 and/or Collins drawing numbers 289-2844 through 2858-00.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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(Project 5955-2012-006)

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